

How Computers Really Work

Why this matters: Every attack, every defense, every piece of malware ultimately runs on hardware. Before you can find vulnerabilities, break into systems, or defend them, you must understand what a computer actually is – a machine built to move electrons around in carefully controlled patterns.

The Zeroes and Ones Are Not Abstract

Inside the machine, everything is voltage. A memory cell holding a **1** is a capacitor holding a charge. A **0** is an empty capacitor. The CPU does not "think" – it switches transistors between conducting and non-conducting states, millions of times per second.

The Five Classic Components

Every computer – a smartphone, a server, a smart thermostat – follows the Von Neumann architecture:

Input – keyboard, mouse, network, sensors

Output – display, network, actuators

Storage – memory and disk

Control unit – the conductor of the orchestra

Arithmetic/Logic Unit (ALU) – does the math

Input → Storage ⇌ Control ⇌ ALU → Output



(all connected by the bus)

The Fetch-Execute Cycle

The CPU runs a relentless three-step loop:

- Fetch** – pull the instruction from memory at the address in the Program Counter
- Decode** – interpret what the instruction means
- Execute** – do it (add, move, branch)

A 3 GHz processor does this roughly 3 billion times per second. This is the clock. Every software exploit ultimately abuses this machinery – usually the memory, the bus, or the privilege levels.

Layers of Abstraction

From hardware up:

Layer	Example
Users & apps	Browser, malware
OS kernel	Windows, Linux
Firmware	UEFI, BIOS, device firmware
Microarchitecture	CPU internals
Digital logic	Gates, register files
Physical	Transistors, electrons

Security insight: Malware can live at *any* layer. Rootkits hide in firmware, bootkits in the boot process, and normal viruses in user applications. The deeper the layer, the harder it is to detect and remove.